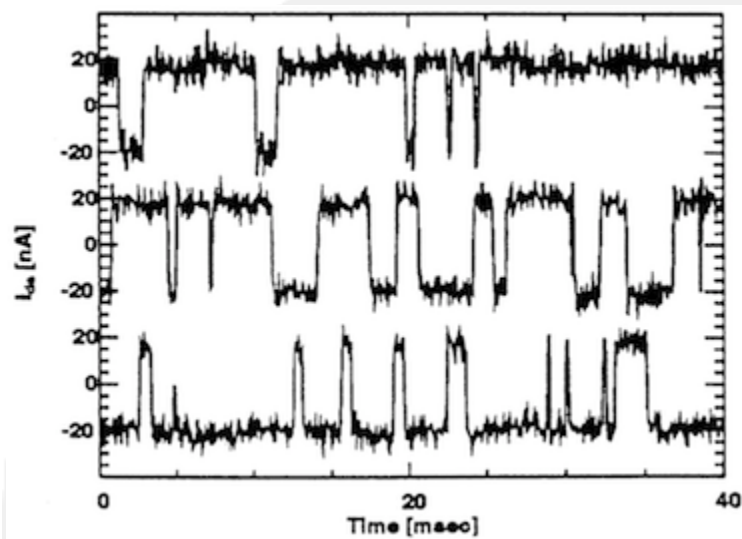


Comprehensive Briefing on Error Control & Hamming Distance

A structured, exam-ready synthesis with conceptual clarity



4	0	1	0	0
---	---	---	---	---

14	1	1	1	0
----	---	---	---	---

$$\text{HammingDistance}(4,14) = 2$$

4	0	1	0	0
---	---	---	---	---

2	0	0	1	0
---	---	---	---	---

$$\text{HammingDistance}(4,2) = 2$$

14	1	1	1	0
----	---	---	---	---

2	0	0	1	0
---	---	---	---	---

$$\text{HammingDistance}(14,2) = 2$$

1 Why Error Control Exists

In any communication system, **noise** may alter transmitted bits.

Error Definition:

Received data $\neq$ Sent data

Error control ensures **data integrity** across unreliable channels.

2 Taxonomy of Transmission Errors

2.1 Single-Bit Error

Only one bit flips ($0 \leftrightarrow 1$).

Example:

Sent: 10110010

Received: 10110000

- Easy to conceptualize
- Rare in high-speed systems

2.2 Burst Error (Most Common in Practice)

Two or more bits change within a data unit.

Burst Length

Distance from **first corrupted bit** to **last corrupted bit**, inclusive.

Even if some bits inside are correct, they count in the burst length.

2.3 Why Burst Errors Dominate

$$\left[\begin{array}{l} \text{Corrupted Bits} = \text{Data Rate} \times \text{Noise Duration} \end{array} \right]$$

Example

Data Rate	Noise Duration	Corrupted Bits
1 Kbps	0.1 sec	100 bits
1 Mbps	0.1 sec	100,000 bits

Modern networks operate at **very high speeds**, so even short noise durations corrupt **many bits**.



Conclusion:

Real-world error control must assume **burst errors**, not isolated bit flips.

3 The Redundancy Principle

Detection and correction both require **extra bits**.

Process

1. Sender generates redundant bits
2. Appends them to data
3. Receiver verifies
4. Redundant bits discarded after validation

Without redundancy → detection is impossible.

4 Detection vs Correction

Feature	Error Detection	Error Correction
Goal	Detect error existence	Fix exact error
Output	Yes / No	Corrected data
Complexity	Low	Very High
Action	Retransmit	No retransmission needed

Why Correction Is Hard

For an n-bit message:

- 1-bit correction → n possibilities
- 2-bit correction → nC_2 possibilities
- k-bit correction → nC_k combinations

For 1000 bits:

$$\left[\binom{1000}{2} \right] = 499,500$$

Correction grows **combinatorially**, detection does not.

5 Error Control Techniques (Layer-Wise View)

Technique	Type	Capability	Layer
Simple Parity	Detection	Detect odd errors	Physical/Data Link
2D Parity	Detection + limited correction	Correct 1-bit	Data Link
Checksum	Detection	Detect many errors	Network/Transport
CRC	Detection	Very strong detection	Data Link
Hamming Code	Correction	Correct 1-bit	Data Link

Practical Insight

- **CRC** → Most important in real networks
- **Hamming Code** → Educationally important, but limited against burst errors

6 Hamming Distance: The Mathematical Foundation

6.1 Definition

Hamming distance between two binary strings =
Number of bit positions where they differ.

6.2 XOR Shortcut

$$[\\ d(x,y) = \text{Number of 1s in } (x \oplus y) \\]$$

Example:

$x = 1011101$
 $y = 1001001$
 $\text{XOR} = 0010100$

Number of 1s = 2
So distance = 2

6.3 Minimum Hamming Distance ($d_{\min}$)

For a set of valid codewords:

$$[\\ d_{\min} = \min d(x,y) \\]$$

If there are n codewords:

$$[\\ \binom{n}{2} \\]$$

pairs must be checked.

7 Core Formulas You Must Memorize

7.1 Error Detection

To detect D-bit errors:

$$\boxed{d_{\min} \geq D + 1}$$

Example:

Detect 10-bit errors $\rightarrow d_{\min} \geq 11$

7.2 Error Correction

To correct D-bit errors:

$$\boxed{d_{\min} \geq 2D + 1}$$

Example:

Correct 5-bit errors $\rightarrow d_{\min} \geq 11$

7.3 Maximum Correctable Errors

$$D = \left\lfloor \frac{d_{\min} - 1}{2} \right\rfloor$$

Example:

If $d_{\min} = 4$

$$D = \left\lfloor \frac{3}{2} \right\rfloor = 1$$

So only 1-bit correction guaranteed.

8 Why the Correction Formula Is $2D + 1$

Imagine spheres around each valid codeword.

To correct D errors:

- Received word must be closer to original than any other codeword.
- That requires **$2D + 1$ separation**.

Geometrically:

- Detection needs gap of $D+1$
- Correction needs separation on both sides

9 Conceptual Hierarchy

Capability	Required $d_{\min}$
Detect 1-bit	2
Detect 2-bit	3
Correct 1-bit	3
Correct 2-bit	5
Correct 3-bit	7